Ayaan Mohammed

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Education

University of Minnesota Twin Cities Minneapolis, MN
Master of Science, Robotics Sep 2025 – Present
National Institute of Technology Puducherry Puducherry, India
Bachelor of Technology, Mechanical Engineering Aug 2019 – April 2023

Experience

Graduate Researcher | Minnesota Interactive Robotics and Vision Lab Feb 2026 – Present

- Working on underwater image restoration for robust downstream robotic tasks such as object detection and SLAM.
- This work is currently in progress for my master's thesis.

Research Assistant | National Institute of Technology Puducherry May 2024 – June 2025

- Designed a YOLOv5-MiDaS pipeline for real-time detection and depth estimation of ghost nets.
- Fine-tuned ResNet50 via transfer learning, achieving 91.1% accuracy on a custom ghost net dataset.
- Benchmarked 3 depth estimation models, identifying MiDaS as optimal for underwater environments.
- Incorporated wavelength-dependent attenuation to refine underwater monocular depth estimation.

Graduate Engineer Trainee | TATA Advanced Systems Ltd. July 2023 – April 2024

- Designed quality control plans for Collins Aerospace Projects ensuring compliance to AS9100.
- Led PPAP and APQP for Collins new product development projects.
- Served on the cross-functional team to optimize manufacturing process of Collins Aerospace Projects.
- Revised CMM inspection plans reducing deviations by 80%.

Research Intern | CSIR-National Institute of Oceanography Goa May 2022 – July 2022

- Generated 3D reconstructions of underwater artifacts using Structure-from-Motion and MVS pipelines.
- Improved point cloud generation by employing colour correction methods.
- Proposed a comprehensive guideline for data acquisition in underwater photogrammetry.

Projects

Spectral Graph Methods for Marine Mammal Vocalization Classification GitHub

- Developed spectral graph classifiers for 4 families and 17 dolphin species.
- Implemented semi-supervised learning using k-NN graphs and harmonic label propagation.
- Catalogued 1,697 recordings across 32 species; extracted MFCCs and mel-spectrograms.
- Achieved 72.47% family-classification accuracy, numerically close to ResNet-18's 73.42%.

Image-Enhancement for ORB-SLAM3 in Underwater Environment GitHub

- Improved ORB-SLAM3 robustness in low-visibility underwater scenes via image enhancement.
- Evaluated image enhancement techniques for underwater ORB-SLAM3 on AQUALOC dataset.
- Achieved 24.7% keyframe tracking improvement using FUnIE-GAN image enhancement.

Class-Aware Seam Carving with Multi-Operator Retargeting GitHub

- Integrated Depth Anything 3 for 3D boundary preservation during class-aware resizing.
- Designed 5 composition-specific energy functions combining classical computer vision with saliency maps to guide seam carving.

Skill Summary

Languages	Python, C/C++, MATLAB
Frameworks	TensorFlow, PyTorch, OpenCV, Open3D, ROS/ROS 2, Gazebo
CAD and 3D	AutoCAD, CATIA, Fusion360, Agisoft Metashape, Meshroom, MeshLab, VisualSfM
Robotics	State Estimation (EKF, UKF), Multi-Sensor Fusion, Visual SLAM, Robotics Perception

Publications

- Ayaan, M., & Raj, R. N. (2024, December). Detection and Monocular Depth Estimation of Ghost Nets. In *International Conference on Computer Vision and Image Processing* (pp. 45–58). Cham: Springer Nature Switzerland.